

33. Generate text embeddings and perform semantic similarity search.

Code:

```
from sentence_transformers import SentenceTransformer
from sklearn.metrics.pairwise import cosine_similarity

# Load the pre-trained embedding model
model = SentenceTransformer("all-MiniLM-L6-v2")

# Documents
documents = [
    "Python is a programming language.",
    "Machine learning is a branch of artificial intelligence.",
    "Artificial intelligence enables computers to perform intelligent tasks.",
    "The sun is the center of our solar system.",
    "Python is commonly used for data science."
]

# Query
query = "What is Python used for?"

# Generate embeddings for documents
document_embeddings = model.encode(documents)

# Generate embedding for query
query_embedding = model.encode([query])

# Calculate cosine similarity
similarity_scores = cosine_similarity(
```

```
    query_embedding,
    document_embeddings
)[0]

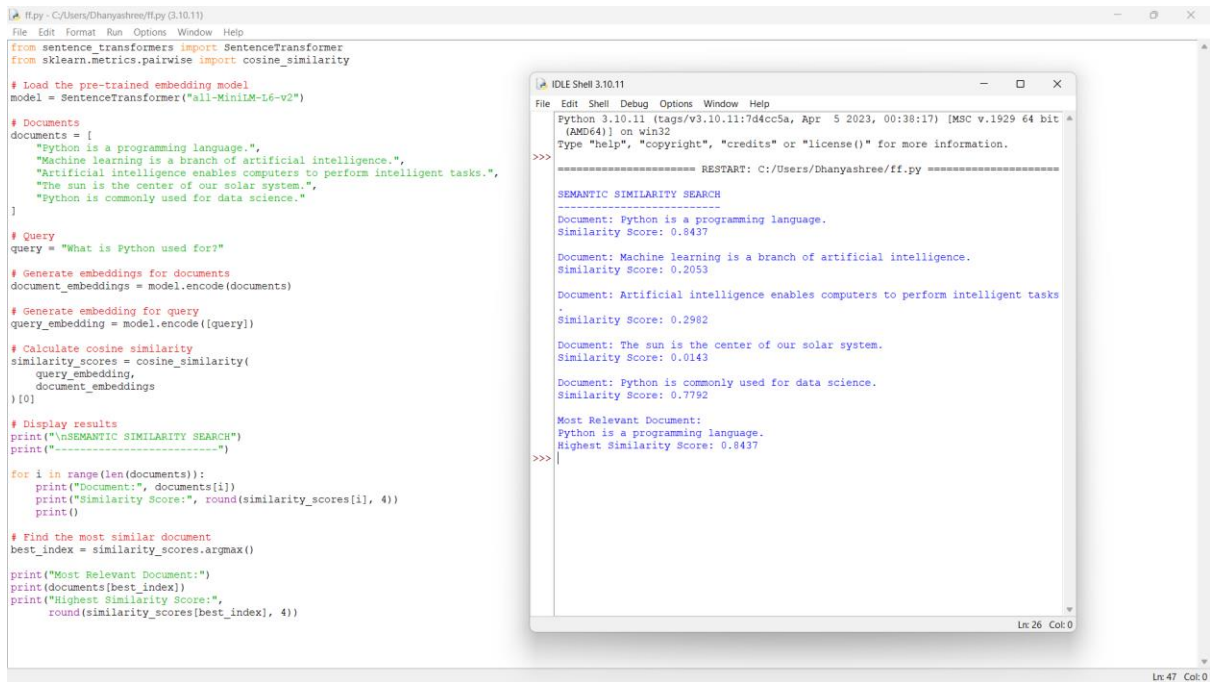
# Display results
print("\nSEMANTIC SIMILARITY SEARCH")
print("-----")

for i in range(len(documents)):
    print("Document:", documents[i])
    print("Similarity Score:", round(similarity_scores[i], 4))
    print()

# Find the most similar document
best_index = similarity_scores.argmax()

print("Most Relevant Document:")
print(documents[best_index])
print("Highest Similarity Score:",
      round(similarity_scores[best_index], 4))
```

output:



The image shows a screenshot of a Python script in an IDE (left) and its output in a terminal window (right).

**Python Script (Left):**

```
ff.py - C:/Users/Dhanyashree/ff.py (3.10.11)
File Edit Format Run Options Window Help
from sentence_transformers import SentenceTransformer
from sklearn.metrics.pairwise import cosine_similarity

# Load the pre-trained embedding model
model = SentenceTransformer("all-MiniLM-L6-v2")

# Documents
documents = [
    "Python is a programming language.",
    "Machine learning is a branch of artificial intelligence.",
    "Artificial intelligence enables computers to perform intelligent tasks.",
    "The sun is the center of our solar system.",
    "Python is commonly used for data science."
]

# Query
query = "What is Python used for?"

# Generate embeddings for documents
document_embeddings = model.encode(documents)

# Generate embedding for query
query_embedding = model.encode([query])

# Calculate cosine similarity
similarity_scores = cosine_similarity(
    query_embedding,
    document_embeddings
)[0]

# Display results
print("\nSEMANTIC SIMILARITY SEARCH")
print("-----")

for i in range(len(documents)):
    print("Document:", documents[i])
    print("Similarity Score:", round(similarity_scores[i], 4))
    print()

# Find the most similar document
best_index = similarity_scores.argmax()

print("Most Relevant Document:")
print(documents[best_index])
print("Highest Similarity Score:",
      round(similarity_scores[best_index], 4))
```

**Terminal Output (Right):**

```
IDLE Shell 3.10.11
File Edit Shell Debug Options Window Help
Python 3.10.11 (tags/v3.10.11:7d4cc5a, Apr 5 2023, 00:38:17) [MSC v.1929 64 bit
(AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/Dhanyashree/ff.py =====
SEMANTIC SIMILARITY SEARCH
-----
Document: Python is a programming language.
Similarity Score: 0.8437
Document: Machine learning is a branch of artificial intelligence.
Similarity Score: 0.2053
Document: Artificial intelligence enables computers to perform intelligent tasks
.
Similarity Score: 0.2982
Document: The sun is the center of our solar system.
Similarity Score: 0.0143
Document: Python is commonly used for data science.
Similarity Score: 0.7792
Most Relevant Document:
Python is a programming language.
Highest Similarity Score: 0.8437
>>>
```

Ln: 26 Col: 0

Ln: 47 Col: 0